In [1]:
```
import geojson
from quantification_utils import (
    ppm_m_to_mol_per_m2,
    ppm_m_to_mol_per_m2_v0,
    calc_L_IME,
    calc_Q_IME,
    reproject_coords
)

from wind_data import get_wind_speed_for_plume,  download_wind_data
from emit_data import get_plumes_data_metadata


bbox = (5.027957, 30.72952, 7.027957, 32.72952)
```

In [2]:
```
def quantify_plume(
    plume_id,
    plume_data,
    plumes_coords,
    lat_plume, long_plume,
    wind_ds,
    source_ds,
    sensing_time: str,
    spatial_resolution: float = 60.0):

    # Get wind data
    u_eff, u_eff_high, u_eff_low, wind_direction, u_wind_component = get_wind_sp
    # Quantify plume
    plume_area, L, IME = calc_L_IME(data=plume_data, plumes_coords=plumes_coords
    emission_rate = calc_Q_IME(u_eff, L, IME)
    emission_rate_low = calc_Q_IME(u_eff_low, L, IME)
    emission_rate_high = calc_Q_IME(u_eff_high, L, IME)

    feature = geojson.Feature(
            geometry=geojson.Point((long_plume, lat_plume)),
            properties={
                "plume_id" : plume_id,
                "Q": emission_rate,
                "Q_low": emission_rate_low,
                "Q_high": emission_rate_high,
                "Q_uncertainty_type": None,
                "wind_direction": wind_direction,
                "u10_wind_speed": u_wind_component,
                "u_eff_wind_speed": u_eff,
                "wind_speed_eff_upper": u_eff_high,
                "wind_speed_eff_lower": u_eff_low,
                "wind_speed_uncertainty_type": None,
                "latitude": lat_plume,
                "longitude": long_plume,
                "IME": IME,
                "length": L,
                "area": plume_area,
                "date": sensing_time.split("T")[0],
            },
        )
    print({
                "plume_id" : plume_id,
                "Q": emission_rate,
```

```
                "Q_low": emission_rate_low,
                "Q_high": emission_rate_high,
                "Q_uncertainty_type": None,
                "wind_direction": wind_direction,
                "u10_wind_speed": u_wind_component,
                "u_eff_wind_speed": u_eff,
                "wind_speed_eff_upper": u_eff_high,
                "wind_speed_eff_lower": u_eff_low,
                "wind_speed_uncertainty_type": None,
                "latitude": lat_plume,
                "longitude": long_plume,
                "IME": IME,
                "length": L,
                "area": plume_area,
                "date": sensing_time.split("T")[0],
            })
    return feature
```

In [3]:

```
## add segmentation to have the real plume not the plume complexe!

def quantify_plumes(
    data_dir= "emit_ch4/EMITL2BCH4PLM",
    sensing_time : str = "20230216",
    bbox = (5.027957, 30.72952, 7.027957, 32.72952),
    spatial_resolution: float = 60.0,
    wind_source=  "GEOS_FP"
) -> list[geojson.Feature] | None:
    """
    Mask the retrieval arary using a threshold and quantify the masked plumes.

    Fields
    ------
    retrieval_array: numpy array with retrieval values
    retrieval_transform: retrieval tiff transfrom
    retrieval_crs: CRS of retrieval tile
    mask: binary mask indicating which pixels are plumes (=1)
    sensing_time: time (str) the sentinel-2 tile was captured
    """
    # Fetch wind tile for date
    wind_ds = download_wind_data(sensing_time , bbox, source=wind_source)
    if wind_ds is None:
        return None

    plumes = get_plumes_data_metadata(base_dir = data_dir , date= sensing_time,
    results = []
    for plume_id, plume in plumes.items():

        # read plume data

        plume_data = plumes[plume_id]['data']
        plume_data[plume_data == -9999] = 0
        plume_data = ppm_m_to_mol_per_m2(plume_data)

        # print(plume_data)
        plume_metadata = plumes[plume_id]['metadata']
        plumes_coords = plumes[plume_id]['geometry']['coordinates'][0]

        print(len(plumes_coords))
        # latitude, longitude
        lat_plume, long_plume = plume_metadata['Latitude of max concentration'],
```

```
        print(lat_plume, long_plume, )
        # quantify one plume
        feature = quantify_plume(plume_id, plume_data, plumes_coords, lat_plume,
                                 wind_ds=wind_ds, source_ds= wind_source,
                                   sensing_time=sensing_time, spatial_resolution
        results.append(feature)

    return results

results = quantify_plumes()
```

```
18
31.74766142315273 5.99925551785579
wind speeeed 4.307325308352157
wind direction -102.12008430857745
L eff 1520.2631351183913
[[5.9946465, 31.749017], [5.9935621, 31.7479325], [5.9892242, 31.7484748], [5.982
1752, 31.7479325], [5.9783796, 31.7446791], [5.9778373, 31.7425102], [5.9783796,
31.7387146], [5.9810907, 31.7360034], [5.9843441, 31.734919], [5.9881398, 31.7354
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[6.0011533, 31.7435947], [6.0033223, 31.7441369], [6.004949, 31.7457636], [6.0044
067, 31.749017], [5.9946465, 31.749017]]
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67.1727380315), (782157.7929286263, 3515996.499061888), (782112.9805470976, 35157
54.549738079), (782175.8996558632, 3515335.004396022), (782441.0943238242, 351504
1.3539360645), (782752.758465067, 3514929.5397148826), (783110.8789223612, 351499
9.5447469116), (784021.0666361015, 3515566.220216791), (784118.8819173579, 351574
9.463809089), (784221.6678861296, 3515752.297347382), (784319.468140976, 3515935.
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017436776), (784611.2043708721, 3516545.4213862796), (783686.2294345736, 3516519.
9171943166)]
origin meters (784127.1708684538, 3516381.6085017966)
max_dist 2214.2356467434793
L from carbon mapper 2214.2356467434793
{'plume_id': '000659', 'Q': np.float64(1625.150574772192), 'Q_low': np.float64(12
26.2528804286596), 'Q_high': np.float64(1858.490767587288), 'Q_uncertainty_type':
None, 'wind_direction': -102.12008430857745, 'u10_wind_speed': -4.21131372451782
2, 'u_eff_wind_speed': np.float64(0.9319046540117077), 'wind_speed_eff_upper': n
p.float64(1.065708139687402), 'wind_speed_eff_lower': np.float64(0.70316608442692
59), 'wind_speed_uncertainty_type': None, 'latitude': 31.74766142315273, 'longitu
de': 5.99925551785579, 'IME': np.float32(66861.01), 'length': np.float64(2214.235
6467434793), 'area': np.float64(2311200.0), 'date': '20230216'}
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31.77802644428709 5.994375425173483
wind speeeed 4.307325308352157
wind direction -102.12008430857745
L eff 2025.8331619360958
[[5.9713305, 31.7853466], [5.9675349, 31.7842621], [5.9637393, 31.7804665], [5.96
15704, 31.7761286], [5.9599437, 31.7701641], [5.9621126, 31.7669107], [5.9659082,
31.7669107], [5.9707883, 31.7696218], [5.972415, 31.7717908], [5.9805485, 31.7717
908], [5.9843441, 31.7728752], [5.9854286, 31.7739597], [5.9919354, 31.7728752],
[5.9978999, 31.7734175], [6.0006111, 31.7755864], [6.0011533, 31.7772131], [6.001
1533, 31.779382], [5.9989844, 31.781551], [5.9973577, 31.781551], [5.9962732, 31.
7826354], [5.9946465, 31.7826354], [5.988682, 31.7853466], [5.9713305, 31.785346
6]]
coords_meters [(781366.48002647, 3520488.4729936663), (781010.2004539844, 352035
8.371525801), (780662.1118561628, 3519927.608078688), (780469.7457040193, 351944
0.91253899), (780333.6411033988, 3518775.2228613035), (780548.9676609838, 351842
0.00143552), (780908.6023016046, 3518429.804904549), (781362.7792129597, 3518743.
103327248), (781510.3289471993, 3518987.8696570043), (782280.944662761, 3519008.9
677238143), (782637.2616031294, 3519139.0996141867), (782736.7096208285, 3519262.
1983784023), (783356.5010874239, 3519158.868963585), (783919.9520062094, 3519234.
5817631194), (784170.1852457725, 3519482.2143077874), (784216.5738543262, 351966
4.044465502), (784209.9322400267, 3519904.590217704), (783997.8152329918, 352013
9.4753734325), (783843.7069750166, 3520135.224197264), (783737.649441696, 352025
2.6584207583), (783583.5431601892, 3520248.411020688), (783010.2250130199, 352053
3.546359611), (781366.48002647, 3520488.4729936663)]
origin meters (783571.9451540393, 3519736.540237777)
max_dist 3377.9793105797453
L from carbon mapper 2500
```

{'plume_id': '000658', 'Q': np.float64(1159.9137069009091), 'Q_low': np.float64(875.209685930366), 'Q_high': np.float64(1326.4548829732066), 'Q_uncertainty_type': None, 'wind_direction': -102.12008430857745, 'u10_wind_speed': -4.211313724517822, 'u_eff_wind_speed': np.float64(0.9319046540117077), 'wind_speed_eff_upper': np.float64(1.065708139687402), 'wind_speed_eff_lower': np.float64(0.7031660844269259), 'wind_speed_uncertainty_type': None, 'latitude': 31.77802644428709, 'longitude': 5.994375425173483, 'IME': np.float32(53879.203), 'length': 2500, 'area': np.float64(4104000.0), 'date': '20230216'}
9
31.78236430444914 6.215606293438081
wind speeeed 4.128521302240252
wind direction -104.9563914479563
L eff 676.1656601750786
[[6.2147929, 31.7831777], [6.2072017, 31.7820932], [6.2061172, 31.7810087], [6.2061172, 31.7788398], [6.2082862, 31.7772131], [6.2158774, 31.7788398], [6.2180463, 31.7810087], [6.2180463, 31.7826354], [6.2147929, 31.7831777]]
coords_meters [(236258.5137962973, 3519781.0146546587), (235536.33013753087, 3519679.197382215), (235430.5058076023, 3519561.568840482), (235424.32387250755, 3519321.042763036), (235625.1645388723, 3519135.366818184), (236348.92410608585, 3519297.3212416163), (236560.54267062567, 3519532.5861620507), (236565.15959190938, 3519712.982714656), (236258.5137962973, 3519781.0146546587)]
origin meters (236333.25351608283, 3519688.8371718)
max_dist 980.5232399529154
L from carbon mapper 980.5232399529154
{'plume_id': '000661', 'Q': np.float64(358.4829472638174), 'Q_low': np.float64(269.1511222687013), 'Q_high': np.float64(410.73871500694526), 'Q_uncertainty_type': None, 'wind_direction': -104.9563914479563, 'u10_wind_speed': -3.988657474517822 3, 'u_eff_wind_speed': np.float64(0.9179133705389472), 'wind_speed_eff_upper': np.float64(1.0517168562146415), 'wind_speed_eff_lower': np.float64(0.6891748009541652), 'wind_speed_uncertainty_type': None, 'latitude': 31.78236430444914, 'longitude': 6.215606293438081, 'IME': np.float32(6630.582), 'length': np.float64(980.5232399529154), 'area': np.float64(457200.0), 'date': '20230216'}
13
31.669579940235813 6.194459225148083
wind speeeed 4.128521302240252
wind direction -104.9563914479563
L eff 844.2748367682174
[[6.1903925, 31.6725622], [6.1871391, 31.6714778], [6.1849702, 31.6693088], [6.1849702, 31.6660554], [6.1865969, 31.6644287], [6.1882236, 31.6638865], [6.1936459, 31.6644287], [6.1968993, 31.6671399], [6.1996104, 31.6676821], [6.2006949, 31.6693088], [6.1996104, 31.6714778], [6.1968993, 31.6725622], [6.1903925, 31.6725622]]
coords_meters [(233630.5281140837, 3507573.496928123), (233318.86988604348, 3507461.1957085202), (233106.95670210867, 3507225.9684132673), (233097.6429655198, 3506865.1756565236), (233247.27801810915, 3506680.7978670965), (233400.01927898388, 3506616.690382715), (233915.87065896782, 3506663.57160621), (234232.18086914817, 3506956.2984743207), (234490.86256309086, 3507009.820123636), (234598.3535135188, 3507187.5744721657), (234501.67269821052, 3507430.7501143496), (234247.63866278407, 3507557.613422187), (233630.5281140837, 3507573.496928123)]
origin meters (234007.71054444654, 3507232.84247003)
max_dist 981.5303785250054
L from carbon mapper 981.5303785250054
{'plume_id': '000660', 'Q': np.float64(814.9732819195726), 'Q_low': np.float64(611.8867720818901), 'Q_high': np.float64(933.7712745769552), 'Q_uncertainty_type': None, 'wind_direction': -104.9563914479563, 'u10_wind_speed': -3.988657474517822 3, 'u_eff_wind_speed': np.float64(0.9179133705389472), 'wind_speed_eff_upper': np.float64(1.0517168562146415), 'wind_speed_eff_lower': np.float64(0.6891748009541652), 'wind_speed_uncertainty_type': None, 'latitude': 31.669579940235813, 'longitude': 6.194459225148083, 'IME': np.float32(15089.414), 'length': np.float64(981.5303785250054), 'area': np.float64(712800.0), 'date': '20230216'}

In [4]:
```
results = quantify_plumes(wind_source="ERA5")
```

```
wind/reanalysis-era5-single-levels_2023-02-16 00:00:00_to_2023-02-16 00:00:00_[3
2.72952, 5.027957, 30.72952, 7.027957].nc
```

```
2025-08-19 16:52:20,465 INFO [2024-09-26T00:00:00] Watch our [Forum](https://foru
m.ecmwf.int/) for Announcements, news and other discussed topics.
```

```
Requesting data with parameters:
{'product_type': 'reanalysis', 'variable': ['10m_u_component_of_wind', '10m_v_com
ponent_of_wind'], 'year': ['2023'], 'month': ['02'], 'day': ['16'], 'time': ['13:
00'], 'format': 'netcdf', 'area': [32.72952, 5.027957, 30.72952, 7.027957]}
18
31.74766142315273 5.99925551785579
wind speeeed 3.5967030823777217
wind direction -101.44632440658208
L eff 1520.2631351183913
[[5.9946465, 31.749017], [5.9935621, 31.7479325], [5.9892242, 31.7484748], [5.982
1752, 31.7479325], [5.9783796, 31.7446791], [5.9778373, 31.7425102], [5.9783796,
31.7387146], [5.9810907, 31.7360034], [5.9843441, 31.734919], [5.9881398, 31.7354
612], [5.9978999, 31.7403413], [5.9989844, 31.741968], [6.0000689, 31.741968],
[6.0011533, 31.7435947], [6.0033223, 31.7441369], [6.004949, 31.7457636], [6.0044
067, 31.749017], [5.9946465, 31.749017]]
coords_meters [(783686.2294345736, 3516519.9171943166), (783586.7707276149, 35163
96.8110695938), (783174.0115523615, 3516445.651086037), (782507.6258255038, 35163
67.1727380315), (782157.7929286263, 3515996.499061888), (782112.9805470976, 35157
54.549738079), (782175.8996558632, 3515335.004396022), (782441.0943238242, 351504
1.3539360645), (782752.758465067, 3514929.5397148826), (783110.8789223612, 351499
9.5447469116), (784021.0666361015, 3515566.220216791), (784118.8819173579, 351574
9.463809089), (784221.6678861296, 3515752.297347382), (784319.468140976, 3515935.
5430079615), (784523.3767877244, 3516001.348866182), (784672.5647852427, 3516186.
017436776), (784611.2043708721, 3516545.4213862796), (783686.2294345736, 3516519.
9171943166)]
origin meters (784127.1708684538, 3516381.6085017966)
max_dist 2214.2356467434793
L from carbon mapper 2214.2356467434793
{'plume_id': '000659', 'Q': np.float64(1521.3904152768057), 'Q_low': np.float64(1
122.4927209332734), 'Q_high': np.float64(1754.730608091902), 'Q_uncertainty_typ
e': None, 'wind_direction': -101.44632440658208, 'u10_wind_speed': -3.52516841888
42773, 'u_eff_wind_speed': np.float64(0.8724058130822746), 'wind_speed_eff_uppe
r': np.float64(1.006209298757969), 'wind_speed_eff_lower': np.float64(0.643667243
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ngitude': 5.99925551785579, 'IME': np.float32(66861.01), 'length': np.float64(221
4.2356467434793), 'area': np.float64(2311200.0), 'date': '20230216'}
23
31.77802644428709 5.994375425173483
wind speeeed 3.5967030823777217
wind direction -101.44632440658208
L eff 2025.8331619360958
[[5.9713305, 31.7853466], [5.9675349, 31.7842621], [5.9637393, 31.7804665], [5.96
15704, 31.7761286], [5.9599437, 31.7701641], [5.9621126, 31.7669107], [5.9659082,
31.7669107], [5.9707883, 31.7696218], [5.972415, 31.7717908], [5.9805485, 31.7717
908], [5.9843441, 31.7728752], [5.9854286, 31.7739597], [5.9919354, 31.7728752],
[5.9978999, 31.7734175], [6.0006111, 31.7755864], [6.0011533, 31.7772131], [6.001
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677238143), (782637.2616031294, 3519139.0996141867), (782736.7096208285, 3519262.
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5817631194), (784170.1852457725, 3519482.2143077874), (784216.5738543262, 351966
4.044465502), (784209.9322400267, 3519904.590217704), (783997.8152329918, 352013
9.4753734325), (783843.7069750166, 3520135.224197264), (783737.649441696, 352025
2.6584207583), (783583.5431601892, 3520248.411020688), (783010.2250130199, 352053
```

```
3.546359611), (781366.48002647, 3520488.4729936663)]
origin meters (783571.9451540393, 3519736.540237777)
max_dist 3377.9793105797453
L from carbon mapper 2500
{'plume_id': '000658', 'Q': np.float64(1085.857288315946), 'Q_low': np.float64(80
1.1532673454028), 'Q_high': np.float64(1252.3984643882434), 'Q_uncertainty_type':
None, 'wind_direction': -101.44632440658208, 'u10_wind_speed': -3.525168418884277
3, 'u_eff_wind_speed': np.float64(0.8724058130822746), 'wind_speed_eff_upper': n
p.float64(1.006209298757969), 'wind_speed_eff_lower': np.float64(0.64366724349749
26), 'wind_speed_uncertainty_type': None, 'latitude': 31.77802644428709, 'longitu
de': 5.994375425173483, 'IME': np.float32(53879.203), 'length': 2500, 'area': np.
float64(4104000.0), 'date': '20230216'}
9
31.78236430444914 6.215606293438081
wind speeeed 3.2538974092986037
wind direction -96.5359723924636
L eff 676.1656601750786
[[6.2147929, 31.7831777], [6.2072017, 31.7820932], [6.2061172, 31.7810087], [6.20
61172, 31.7788398], [6.2082862, 31.7772131], [6.2158774, 31.7788398], [6.2180463,
31.7810087], [6.2180463, 31.7826354], [6.2147929, 31.7831777]]
coords_meters [(236258.5137962973, 3519781.0146546587), (235536.33013753087, 3519
679.197382215), (235430.5058076023, 3519561.568840482), (235424.32387250755, 3519
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297.3212416163), (236560.54267062567, 3519532.5861620507), (236565.15959190938, 3
519712.982714656), (236258.5137962973, 3519781.0146546587)]
origin meters (236333.25351608283, 3519688.8371718)
max_dist 980.5232399529154
L from carbon mapper 980.5232399529154
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max_dist 981.5303785250054
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In [5]:

```python
import rasterio
import numpy as np

def extract_plume_coords(tif_file, threshold=0):
    """
    Extract (x, y) coordinates of plume pixels from a GeoTIFF.

    Parameters
    ----------
    tif_file : str
        Path to plume GeoTIFF (concentration map).
    threshold : float
        Minimum value to consider as plume (e.g., > 0, or a ppm threshold).

    Returns
    -------
    coords : list of tuples
        List of (x, y) coordinates in the GeoTIFF CRS.
    crs : rasterio.crs.CRS
        CRS of the coordinates (useful to reproject to meters).
    """
    with rasterio.open(tif_file) as src:
        data = src.read(1)  # first band
        transform = src.transform
        crs = src.crs

        # Find plume pixels (mask)
        plume_mask = data > threshold

        # Get row, col indices of plume pixels
        rows, cols = np.where(plume_mask)

        # Convert to (x, y) coordinates
        coords = [rasterio.transform.xy(transform, r, c) for r, c in zip(rows, c

    return data, coords, crs


def quantify_plume_carbon_mapper(
    data_file= "carbon_mapper/emi20230216t133626p09013-A_l3a-ch4-mfa-v0_concentr
    sensing_time : str = "20230216",
    long_plume  =  5.99925551785579,
    lat_plume = 31.74766142315273,
    spatial_resolution: float = 60.0,
    plume_id =  "000659",
    wind_source=  "ERA5",
    bbox = (5.027957, 30.72952, 7.027957, 32.72952),

) -> list[geojson.Feature] | None:
    """
    Mask the retrieval arary using a threshold and quantify the masked plumes.

    Fields
    ------
    retrieval_array: numpy array with retrieval values
    retrieval_transform: retrieval tiff transfrom
```

```
    retrieval_crs: CRS of retrieval tile
    mask: binary mask indicating which pixels are plumes (=1)
    sensing_time: time (str) the sentinel-2 tile was captured
    """
    # Fetch wind tile for date
    wind_ds = download_wind_data(sensing_time , bbox, source=wind_source)
    if wind_ds is None:
        return None

    # Load raster
    plume_data, coords, crs = extract_plume_coords(data_file, threshold=0)
    print(coords)
    plume_data = ppm_m_to_mol_per_m2(plume_data)

    # read plume data
    print(lat_plume, long_plume, plume_data )
    # quantify one plume
    feature = quantify_plume(plume_id, plume_data, plumes_coords = coords,
                             long_plume =long_plume , lat_plume= lat_plume, wind
                             sensing_time=sensing_time, spatial_resolution=spati

    return feature

result = quantify_plume_carbon_mapper(data_file = "carbon_mapper/emi20230216t133
result
```

```
wind/reanalysis-era5-single-levels_2023-02-16 00:00:00_to_2023-02-16 00:00:00_[3
2.72952, 5.027957, 30.72952, 7.027957].nc
```

```
2025-08-19 16:52:20,741 INFO [2024-09-26T00:00:00] Watch our [Forum](https://foru
m.ecmwf.int/) for Announcements, news and other discussed topics.
```

```
Requesting data with parameters:
{'product_type': 'reanalysis', 'variable': ['10m_u_component_of_wind', '10m_v_com
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Out[5]:


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In [6]:

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max_dist 551616.9095253561
L from carbon mapper 2500
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de': 5.99925551785579, 'IME': np.float64(20855.028489417895), 'length': 2500, 'ar
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```

Out[6]:

```
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```


In [7]:
```
(4.21 + 3.52) /2
```

Out[7]:
```
3.865
```

In [8]:
```
import rasterio
import matplotlib.pyplot as plt
import json
import geopandas as gpd
from rasterio.plot import show
from shapely.geometry import box


def visualize_plume_complex(tif_path):
    # --- Load raster ---
    with rasterio.open(tif_path) as src:
        data = src.read(1)  # first band
        print(data)
        bounds = src.bounds
        crs = src.crs
        plt.figure(figsize=(6, 5))
        plt.imshow(data, cmap="viridis")
        plt.colorbar(label="CH₄ Enhancement")
        plt.title(f"Plume Complex")
```

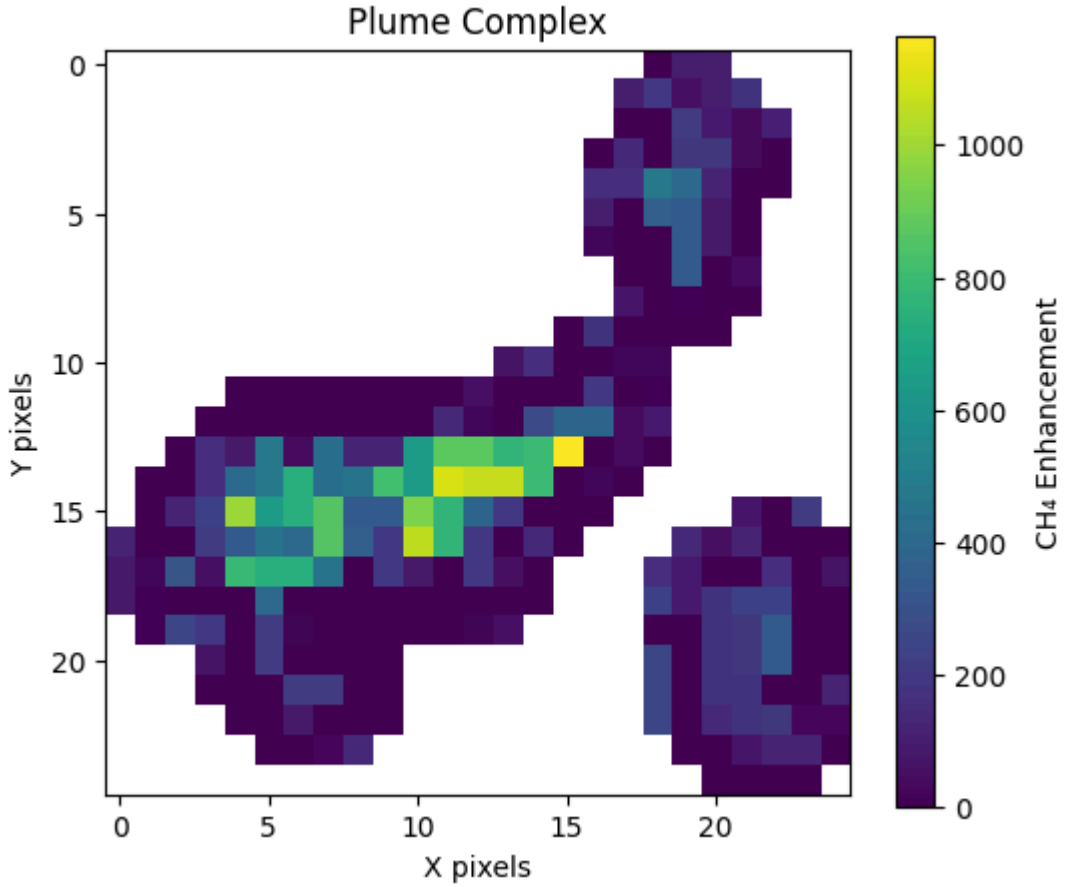
```
        plt.xlabel("X pixels")
        plt.ylabel("Y pixels")
        plt.show()


visualize_plume_complex("carbon_mapper/emi20230216t133626p09013-D_l3a-ch4-mfa-v0
```

```
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             nan            nan            nan            nan
             nan            nan            nan            nan
             nan            nan            nan            nan
             nan            nan 0.00000000e+00 9.62054032e+01
  9.99957036e+01            nan            nan            nan
             nan]
 [           nan            nan            nan            nan
             nan            nan            nan            nan
             nan            nan            nan            nan
             nan            nan            nan            nan
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  9.99957036e+01 1.82375400e+02            nan            nan
             nan]
 [           nan            nan            nan            nan
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             nan            nan            nan            nan
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  9.52253599e+01 3.34384231e+01 1.10356788e+02            nan
             nan]
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             nan            nan            nan            nan
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             nan]
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  1.17471373e+02 0.00000000e+00 0.00000000e+00            nan
             nan]
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             nan]
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             nan            nan            nan            nan
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  8.29182956e+01 0.00000000e+00            nan            nan
             nan]
 [           nan            nan            nan            nan
             nan            nan            nan            nan
             nan            nan            nan            nan
             nan            nan            nan            nan
             nan 0.00000000e+00 0.00000000e+00 3.37031332e+02
  0.00000000e+00 3.94639001e+01            nan            nan
             nan]
 [           nan            nan            nan            nan
             nan            nan            nan            nan
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             nan            nan            nan            nan
```

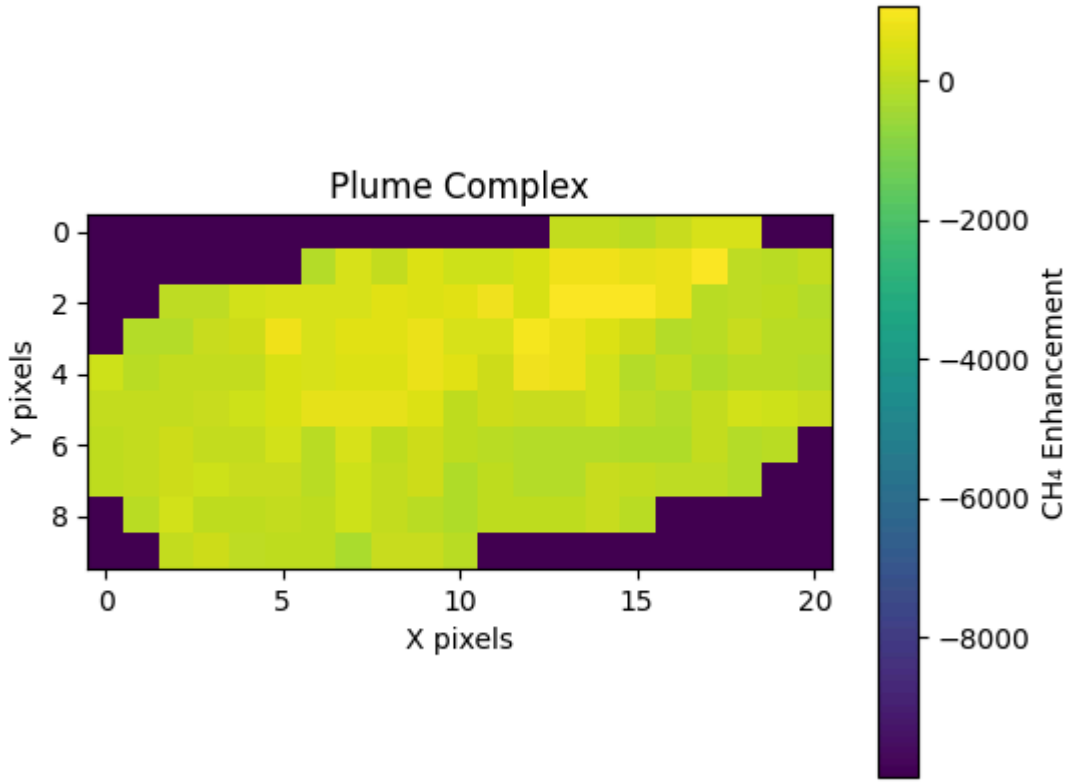
```
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            nan]
[           nan             nan             nan             nan
            nan             nan             nan             nan
            nan             nan             nan             nan
            nan             nan             nan 0.00000000e+00
 1.75542343e+02 0.00000000e+00 0.00000000e+00 0.00000000e+00
 0.00000000e+00             nan             nan             nan
            nan]
[           nan             nan             nan             nan
            nan             nan             nan             nan
            nan             nan             nan             nan
            nan 6.06429426e+01 1.60170863e+02 0.00000000e+00
 0.00000000e+00 2.72027248e+01 2.72027248e+01             nan
            nan             nan             nan             nan
            nan]
[           nan             nan             nan             nan
 0.00000000e+00 0.00000000e+00 0.00000000e+00 0.00000000e+00
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            nan]
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 2.00985077e+01 0.00000000e+00 2.80830241e+02 3.80284404e+02
 3.80284404e+02 4.50496055e+01 9.26389399e+01             nan
            nan             nan             nan             nan
            nan]
[           nan             nan 0.00000000e+00 1.65017186e+02
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 1.31252271e+02 1.31252271e+02 6.55657025e+02 8.74343767e+02
 8.74343767e+02 7.67512953e+02 8.04345622e+02 1.16351088e+03
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            nan]
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 4.46311392e+02 8.15934053e+02 6.55657025e+02 1.09597820e+03
 1.07108359e+03 1.07108359e+03 8.04345622e+02 0.00000000e+00
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 0.00000000e+00]
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```

```
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 6.24287195e+01]
[8.28147563e+01 8.54183805e+00 0.00000000e+00 0.00000000e+00
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            nan            nan 2.35169463e+02 9.46974349e+01
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 0.00000000e+00]
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 0.00000000e+00]
[           nan            nan            nan 6.28874567e+01
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            nan            nan            nan            nan
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 0.00000000e+00]
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 1.18311571e+02]
[           nan            nan            nan            nan
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            nan            nan            nan            nan
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[           nan            nan            nan            nan
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            nan            nan            nan            nan
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```



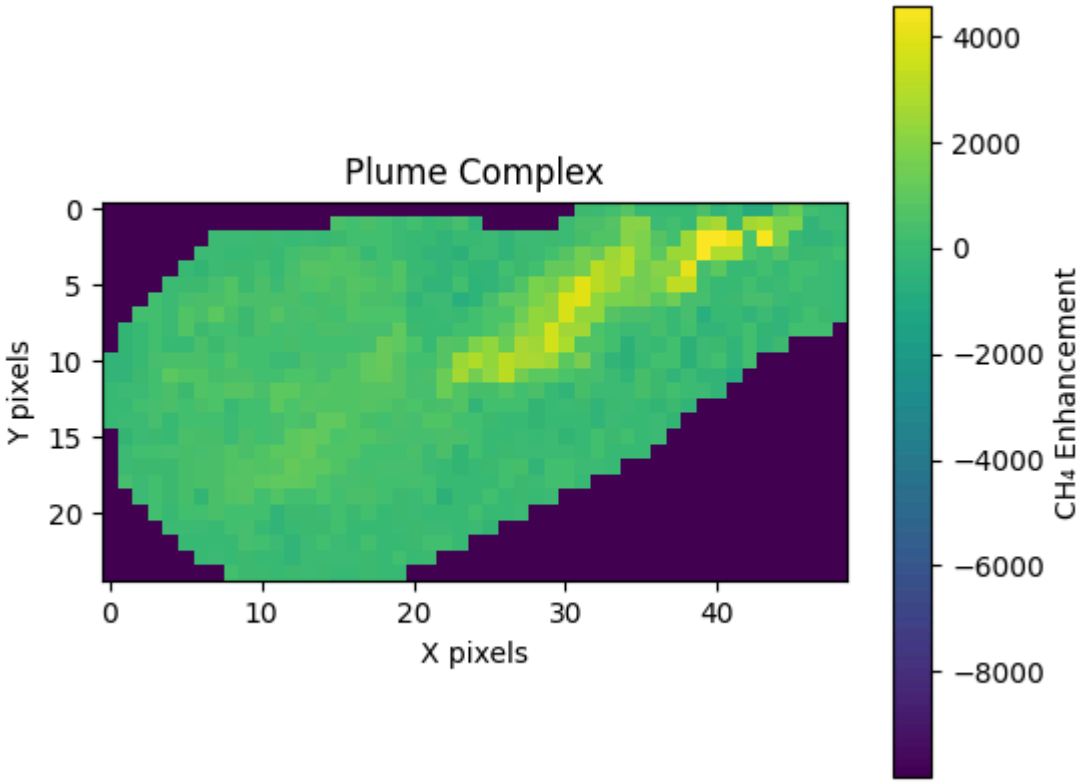
In [9]:
```
visualize_plume_complex("emit_ch4/EMITL2BCH4PLM/EMIT_L2B_CH4PLM_001_20230216T133
```

```
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  -9.99900000e+03 -9.99900000e+03 -9.99900000e+03 -9.99900000e+03
  -9.99900000e+03  1.27179222e+02  1.60237244e+02 -2.70161648e+01
   1.84448410e+02  3.82126953e+02  3.82126953e+02 -9.99900000e+03
  -9.99900000e+03]
 [-9.99900000e+03 -9.99900000e+03 -9.99900000e+03 -9.99900000e+03
  -9.99900000e+03 -9.99900000e+03 -1.43562180e+02  3.81635437e+02
   1.09240425e+02  4.78423187e+02  2.83996948e+02  2.83996948e+02
   4.62152466e+02  8.21341187e+02  8.21341187e+02  6.59596985e+02
   7.44785889e+02  1.07173682e+03  1.13642292e+01 -6.93423080e+01
   1.31181808e+02]
 [-9.99900000e+03 -9.99900000e+03  2.43280869e+01  2.43280869e+01
   2.95344391e+02  4.13281677e+02  4.13281677e+02  3.81635437e+02
   6.25016479e+02  4.78423187e+02  5.76075867e+02  8.78896057e+02
   4.62152466e+02  1.04373547e+03  1.03907776e+03  1.03907776e+03
   7.44785889e+02 -6.26743431e+01  1.13642292e+01  4.96075821e+01
  -1.40800613e+02]
 [-9.99900000e+03 -1.17418564e+02 -1.17418564e+02  1.79907059e+02
   2.26810196e+02  8.49184998e+02  4.10282654e+02  5.90256531e+02
   6.25016479e+02  8.09279419e+02  3.82541534e+02  3.82541534e+02
   9.54267456e+02  7.61163879e+02  5.39998596e+02  2.18117218e+02
  -1.42810028e+02 -6.26743431e+01  1.75712631e+02 -3.41323090e+01
  -3.41323090e+01]
 [ 2.89552551e+02 -2.37645912e+01  1.47735870e+02  1.47735870e+02
   1.56917557e+02  4.49585632e+02  4.10282654e+02  5.21858032e+02
   4.81721069e+02  8.09279419e+02  5.84476440e+02  2.43506027e+02
   8.93176025e+02  7.61163879e+02  3.44680115e+02 -1.35463486e+02
   9.52508316e+01 -2.05707260e+02 -3.01853237e+01 -3.01853237e+01
  -1.02160881e+02]
 [ 9.89819107e+01  1.46000397e+02  1.02918427e+02  1.76552307e+02
   2.71488068e+02  4.49585632e+02  6.89057129e+02  6.80464905e+02
   6.80464905e+02  4.88097046e+02  7.68504181e+01  2.43506027e+02
   1.98075577e+02  1.70640915e+02  3.44680115e+02  1.95657978e+01
  -9.78991241e+01  1.00571533e+02  3.04342255e+02  2.63598389e+02
   1.93437317e+02]
 [ 5.63302193e+01  1.46000397e+02  2.33424561e+02  8.48999634e+01
   8.48999634e+01  2.97632996e+02 -2.46595955e+01  4.05442841e+02
   6.80296249e+01  2.28792252e+02  7.68504181e+01 -7.45352402e+01
  -1.81483093e+02 -1.81483093e+02 -1.39292877e+02 -2.25314423e+02
  -2.25314423e+02  1.00571533e+02 -1.63897842e+02 -9.35817642e+01
  -9.99900000e+03]
 [ 5.63302193e+01  1.00977478e+02  2.33424561e+02  2.75210327e+02
   1.98150238e+02  1.98150238e+02 -2.46595955e+01  2.72408630e+02
   1.38033173e+02  2.28792252e+02 -2.11381271e+02  4.27596092e+01
  -1.19234612e+02 -1.67655746e+02  1.70415634e+02  1.34767624e+02
   9.19409320e-02  2.32529564e+01 -1.63897842e+02 -9.99900000e+03
  -9.99900000e+03]
 [-9.99900000e+03 -4.39058800e+01  3.74755859e+02  7.09129410e+01
   7.09129410e+01  1.45399277e+02  5.91211739e+01  2.72408630e+02
   1.11199554e+02 -5.35133743e+01 -2.11381271e+02  4.27596092e+01
   3.69947357e+01  3.69947357e+01  1.70415634e+02 -8.07603531e+01
  -9.99900000e+03 -9.99900000e+03 -9.99900000e+03 -9.99900000e+03
  -9.99900000e+03]
 [-9.99900000e+03 -9.99900000e+03  1.46983780e+02  2.44989258e+02
   4.87491876e-01  6.77723007e+01  5.91211739e+01 -2.75265289e+02
   1.75939194e+02  1.75939194e+02 -7.73341522e+01 -9.99900000e+03
  -9.99900000e+03 -9.99900000e+03 -9.99900000e+03 -9.99900000e+03
  -9.99900000e+03 -9.99900000e+03 -9.99900000e+03 -9.99900000e+03
  -9.99900000e+03]]
```



In [10]:
```
visualize_plume_complex("emit_ch4/EMITL2BCH4PLM/EMIT_L2B_CH4PLM_001_20230216T133
```

```
[[-9999.       -9999.       -9999.       ...   211.07219     17.476784
    -83.55493 ]
 [-9999.       -9999.       -9999.       ...  -421.75537     72.769585
     72.769585]
 [-9999.       -9999.       -9999.       ...  -421.75537   -362.13153
   -196.48773 ]
 ...
 [-9999.       -9999.       -9999.       ... -9999.       -9999.
  -9999.      ]
 [-9999.       -9999.       -9999.       ... -9999.       -9999.
  -9999.      ]
 [-9999.       -9999.       -9999.       ... -9999.       -9999.
  -9999.      ]]
```


Plume Complex
Y pixels
X pixels
0
5
10
15
20
0
10
20
30
40
4000
2000
0
−2000
−4000
−6000
−8000
CH₄ Enhancement

In [ ]: